Project

Google Stock Data Analysis

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Objective

The objective of this project is to analyse Google stock data to identify price trends and patterns over time. It aims to help investors make informed decisions by visualizing stock behaviour. The analysis will also provide insights into future stock price movements. Ultimately, the goal is to support smarter investment strategies.

Data Description

The data used in this project is sourced from Google Sheets, which provides historical Google stock data. The dataset is extracted using formulas within Google Sheets to automate the process of fetching the stock data and ensuring it is up-to-date.

=GOOGLEFINANCE("GOOG", "all", DATE(2022,1,1), DATE(2024,12,31), "DAILY")

Data Description:

The dataset includes the following columns:

- Date
- Opening Price
- High Price
- Low Price
- Closing Price
- Volume

Data Models

Entities and Attributes:

1. Entity: Google Stock Data
    - Attributes:
        - Date: The date when the stock was traded (e.g., "01-01-2024").
        - Open Price: The price of the stock at the opening of the market for that day.
        - High Price: The highest price the stock reached during the day.
        - Low Price: The lowest price the stock reached during the day.
        - Close Price: The price of the stock at the market’s close for that day.
        - Volume: The number of shares traded on that day.

Approach

1. Data Collection: Import Google stock data from Google Sheets, which includes daily trading information (Open, Close, High, Low, Volume).
2. Import Libraries: Pandas, NumPy, Matplotlib, Seaborn.
3. Data Cleaning: Handle missing or incorrect values, ensure date format consistency, and preprocess data for analysis.
4. Data Aggregation:
    - Group the data by year and month to calculate yearly and monthly summaries (e.g., average close price, total volume).
    - Calculate key metrics like moving averages.
5. Visualization:
    - Create various visualizations like line charts, bar charts, scatter plots, pie charts, etc., to represent stock performance, volume trends, and changes.
    - Generate yearly and monthly visualizations for a clearer understanding of the stock's behavior.
6. Conclusion: Summarize the findings and provide recommendations for investors based on the analysis.

Conclusion

This project analyses Google stock data to understand how its price changes over time. Visualization makes the data easier to understand, showing trends and patterns clearly. Investors, analysts, and business people can use this analysis to make better decisions about buying or selling stocks. By looking at past performance, they can predict future price movements and avoid risks. This helps people make smarter choices when investing. The analysis also provides valuable insights into stock behavior, which can guide future investment strategies. Overall, it helps users plan and manage their investments more effectively.